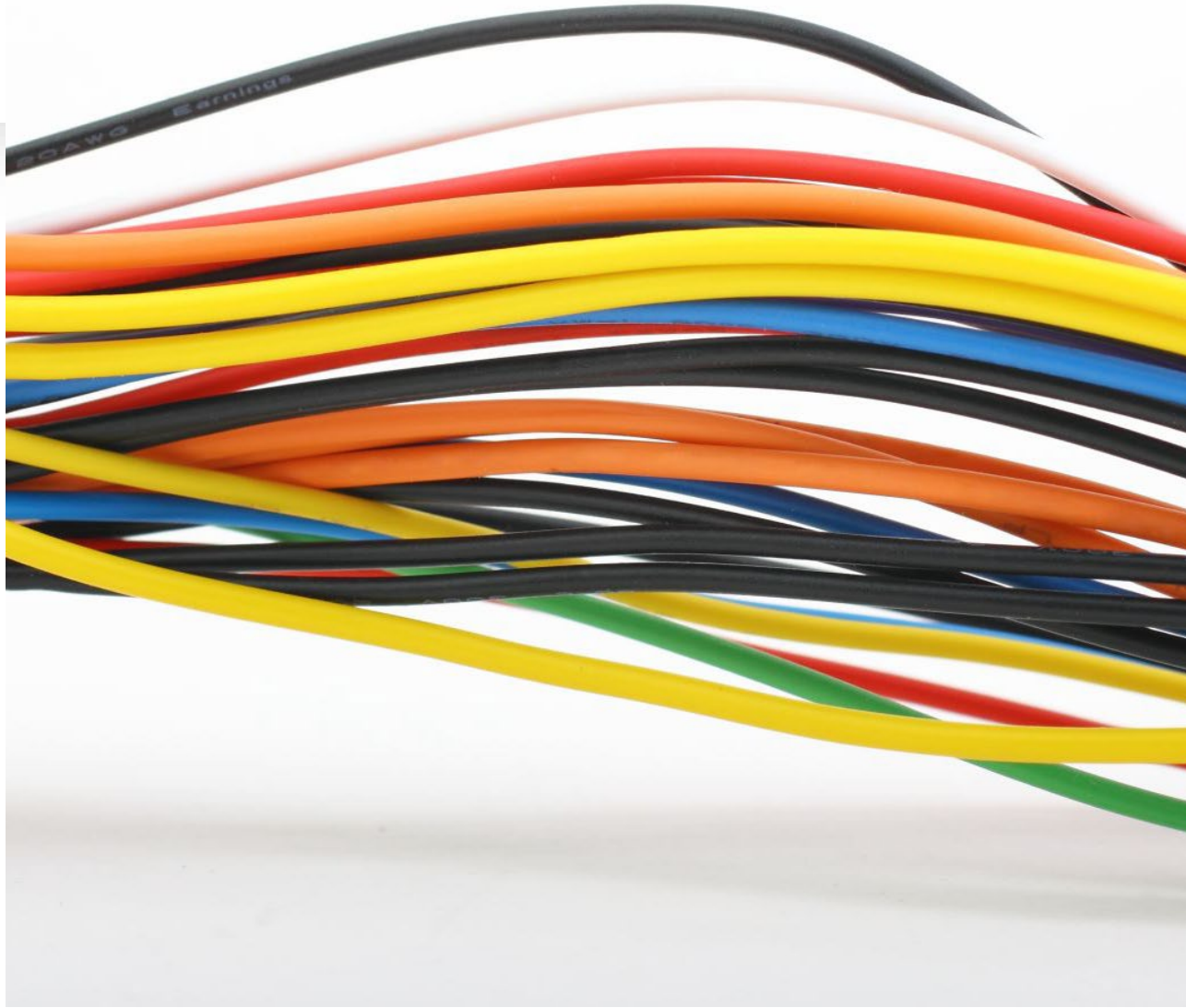


CSCI368: Network Security – Lab 4

HTTP Request Interception, Traffic Analysis, and Modification Using Burp Suite & Wireshark

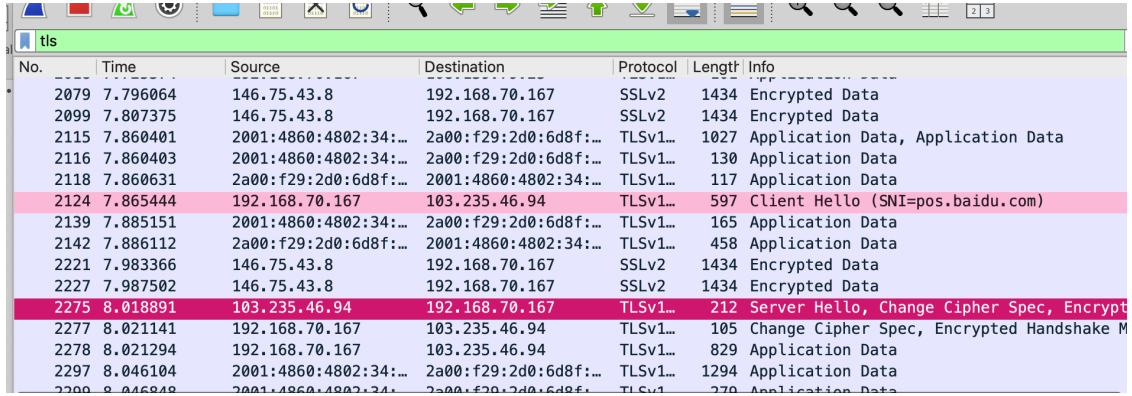
Lab Overview

- Explore web traffic analysis using Wireshark and Burp Suite.
- Understand how to intercept, modify, and reissue HTTP/HTTPS requests.
- **Tools:**
  - Wireshark
  - Burp Suite



Wireshark for Understanding Networks & Identifying Weaknesses

- Wireshark is an essential tool for network analysis and cybersecurity. It allows you to
  - **Capture and analyze traffic** and understand how data moves across the network.
  - **Identify potential vulnerabilities** and discover unencrypted (HTTP) traffic, insecure protocols, or misconfigured security settings.
  - **Monitor TLS Handshake** to ensure encrypted connections are secure and correctly implemented.
- Handshake Steps:
  - Client Hello: The client proposes encryption methods and starts the secure session.
  - Server Hello: The server responds with a certificate and chosen encryption method.
- Why it matters: Any issues in this process (like weak ciphers or improper certificate validation) could indicate vulnerabilities such as susceptibility to MiTM attacks.



Part 1 – Analyzing Web Traffic with Wireshark



Launch Wireshark, select a network interface.



Capture traffic, filter HTTP/HTTPS.



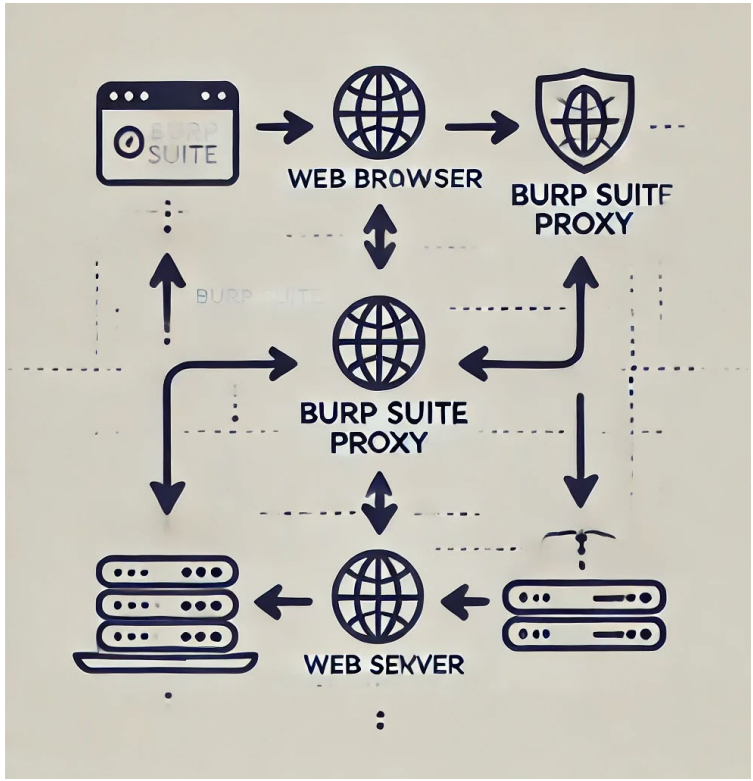
Observe TLS/SSL handshake and encrypted data.



Stop and save the capture.

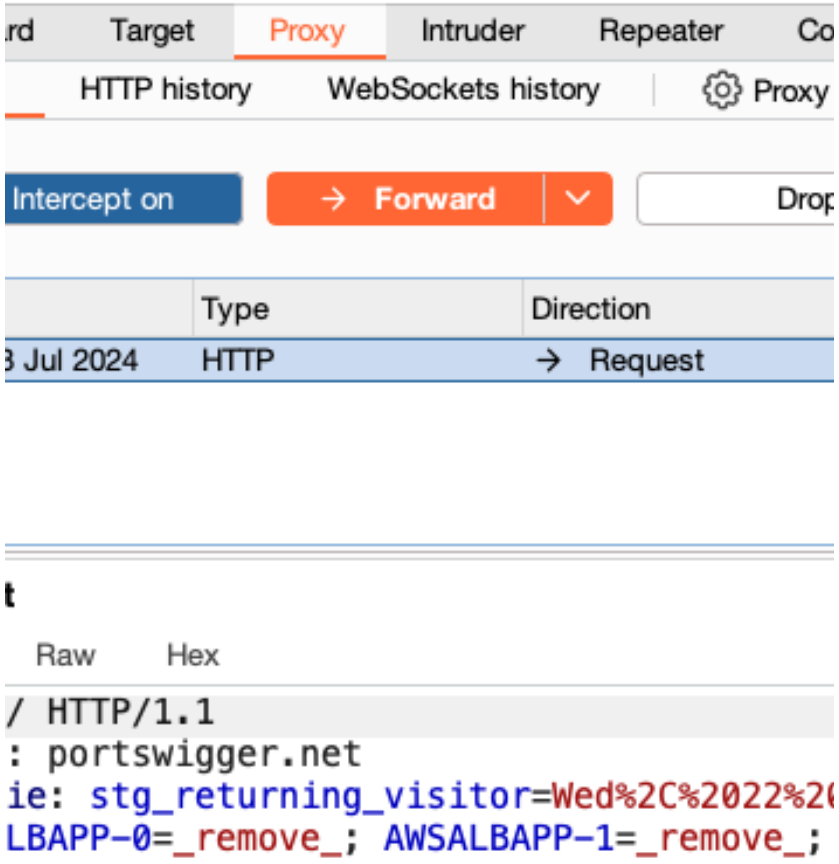
Burp Suite for Network Traffic Capture & Interception

- Burp Suite is a powerful web security testing tool that helps analyze, intercept, and manipulate web traffic. Penetration testers and cybersecurity professionals primarily use it to:
  - **Capture web traffic**: Monitor requests and responses between browsers and web servers.
  - **Analyze vulnerabilities**: Identify weaknesses like improper data handling, security misconfigurations, or injection points.
  - **Modify traffic**: Intercept and alter data sent between client and server to simulate attacks or test security measures.
- **How Burp Suite Helps in Network Traffic Capture**
  - **Traffic Interception**:
    - Burp Suite acts as a **proxy** between your browser and the internet.
    - It captures **HTTP/HTTPS traffic** in real time, enabling you to inspect and modify requests before they reach the server or before responses are delivered to the browser.
  - **Proxy Mode**:
    - Burp Suite can be configured to work as a proxy server, intercepting all web traffic.



Intercepting Traffic

- **What Does Traffic Interception Mean?**
  - **Intercepting Traffic** temporarily halts the data flow between the client (e.g., web browser) and the server.
  - For example, you can **pause** a request, examine it, and **modify** its content before allowing it to continue.
- **Why do this?**
  - **Security Testing**: Test how a web application handles manipulated data (e.g., by changing parameters in an intercepted HTTP request).
  - **Exploit Discovery**: By intercepting traffic, you can identify hidden vulnerabilities like insecure data transmission, logic flaws, or weak validation checks.
  - **Testing Responses**: Simulate different responses from the server by modifying requests and observing how the server reacts.



Part 2 – HTTP Request Interception with Burp Suite



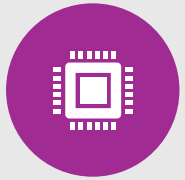
Set up Burp Suite (Proxy > Intercept).



Open Burp's browser and visit Google.com.



Intercept, view, and forward HTTP requests.



Review HTTP history to analyze requests.

Modifying HTTP Requests

- **What does it mean to modify HTTP requests?**
  - HTTP requests are messages sent by a client (e.g., browser) to a server when interacting with a web page.
  - Modifying these requests involves intercepting the original request and changing key parameters before sending it to the server.
- **Why Modify HTTP Requests?**
  - **Testing Security**: Developers or security testers may alter requests to test how a server handles unexpected inputs.
  - **Simulating Attacks**: By changing parameters (e.g., price, product ID), testers can simulate attacks like price manipulation or authentication bypass.
  - **Exploring Application Logic**: Helps understand how the server processes requests and detects vulnerabilities like logic flaws.

Part 3 – Modifying HTTP Requests in Burp Proxy



LOG INTO A SHOPPING SITE.



ADD AN ITEM TO THE CART AND INTERCEPT THE POST REQUEST.



MODIFY THE PRICE IN THE INTERCEPTED REQUEST.

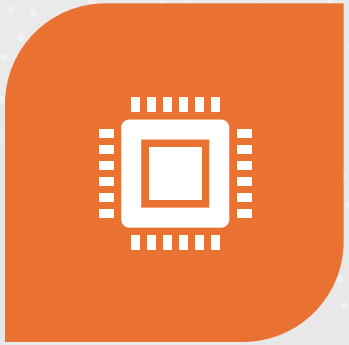


VERIFY THE PRICE MODIFICATION.

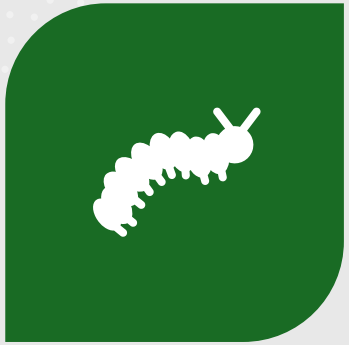
Reissuing Requests with Burp Repeater

- **What does it mean to Reissue HTTP Requests?**
  - Reissuing a request involves sending the same HTTP request multiple times, often with modifications, to observe how the server responds.
  - This helps testers explore how a web server handles different inputs without reloading or interacting with the browser.
- **Purpose of Reissuing Requests:**
  - **Test Input Variations**: Modify parameters such as IDs, query strings, or form data to see how the server reacts to each variation.
  - **Identify Vulnerabilities**: This helps identify how the server handles incorrect or malicious inputs, such as SQL injection, buffer overflows, or invalid data.
  - **Error Handling**: Reveals potential information leaks, such as debugging messages or stack traces, when an unexpected input causes server errors.
  - **Replay Attacks**: Test if an application is vulnerable to repeated requests that might exploit session handling.
- **Using Burp Repeater**:
  - Burp Repeater allows you to resend captured requests, modify them, and view server responses in a controlled environment.
  - You can analyze the server's behavior without affecting your web page or application interaction.

Part 4 – Reissuing Requests with Burp Repeater



IDENTIFY A REQUEST FROM HTTP HISTORY.



SEND THE REQUEST TO BURP REPEATER.



MODIFY PARAMETERS AND OBSERVE SERVER RESPONSES.

Man-in-the-Middle (MiTM) Attack

- **What is a MiTM Attack?**
  - A **Man-in-the-Middle (MiTM) attack** occurs when an attacker secretly intercepts and possibly alters the communication between two parties who believe they are directly communicating.
  - The attacker can eavesdrop, capture sensitive information (passwords, session tokens), or modify messages before sending them.
- **How is it Done?**
  - **Intercepting Communication**:
    - The attacker places themselves between the client (e.g., browser) and server (e.g., web application).
    - They may use tools like Burp Suite to intercept unencrypted or encrypted traffic (with SSL stripping techniques).
  - **Decrypting SSL/TLS Traffic**:
    - Attackers install rogue SSL certificates (like Burp's CA certificate) to decrypt HTTPS traffic.
    - This allows attackers to view encrypted data as plain text and potentially alter it.
  - **Altering Communication**:
    - After intercepting the traffic, attackers can modify HTTP requests and responses before forwarding them to the original destination.
    - Example: Change the amount in a transaction or manipulate login details.

Man-in-the-Middle (MiTM) Attack

- **Example of MiTM Attack**:
  - A hacker intercepts the communication between a user’s browser and an online bank.
  - The attacker can capture login credentials and modify transactions before they reach the bank's server.
- **Why is MiTM Dangerous?**
  - **Data Theft**: Sensitive data such as passwords, credit card numbers, or private conversations can be stolen.
  - **Undetected Modifications**: The attacker's changes are hard to detect, making them a covert and effective form of attack.
- **Defense Mechanisms**:
  - **Encryption (SSL/TLS)**: Ensures data is encrypted during transmission, making it harder for attackers to interpret.
  - **Certificates and Public Key Infrastructure (PKI)**: Only trusted parties can participate in secure communication.
  - **VPNs**: Secures data at the network level, preventing unauthorized access.

Submission Instructions



Wireshark Capture File:
Submit the .pcap file.



Burp Suite Screenshots:
Submit screenshots of intercepted and modified requests.

Conclusion



Web Traffic Analysis: Tools like Wireshark and Burp Suite help analyze network traffic and detect vulnerabilities.



Interception and Modification: Burp Suite enables traffic interception and modification, useful for testing web security.



Regularly analyzing and securing web traffic is crucial to defending against attacks.